Risk-Controlled Spatio-Temporal Graph Learning for Anti-Money Laundering Under Extreme Imbalance and Topological Camouflage

Md. Nazmul, Musrat Jahan Gungun, and Maheli Ahmed

Abstract—**Financial money laundering moves an estimated \$800B–\$2T annually across global financial rails. While Graph Neural Networks (GNNs) capture relational topologies, practical Anti-Money Laundering (AML) surveillance is hindered by velocity evasion, topological camouflage, extreme class imbalance (π < 0.05%), cold-start isolation, and uncalibrated uncertainty. We propose C-STGB (Conformal Spatio-Temporal GraphBoost), an integrated risk-controlled temporal graph learning framework combining continuous harmonic temporal attention, learnable edge-trust gating, typology-clustered latent GraphSMOTE, evidence-adaptive graph-tabular fusion, and Class-Conditional Conformal Risk Control (CRC). Benchmarked across 14 financial transaction networks (9.53M entities, 9.53M transactions, 182 evaluation pairs over five locked seeds), C-STGB achieves an unweighted macro F1 of 67.26% (0.6848 PR-AUC). This establishes parity with tuned gradient-boosted decision trees (XGBoost 67.92%, p = 0.8077; CatBoost 67.14%, p = 0.4318) which excel on flat transaction streams, alongside statistically significant outperformance over evaluated deep GNNs (+43.49 pp mean uplift over GCN; W = 105.0, p_{adj} < 0.001), with advantages concentrated in graph-rich relational regimes. Furthermore, C-STGB provides finite-sample coverage (≥ 99.0%) under within-class exchangeability with Adaptive Conformal Inference (ACI) drift tracking, routing > 99.4% of volume into decisive singleton prediction sets while achieving a streaming Fast-Path inference latency of 0.45 ms per event.**

Index Terms—**Anti-Money Laundering, Graph Neural Networks, Temporal Graph Learning, Conformal Prediction, Financial Forensics.**

I. Introduction

FINANCIAL crime surveillance is a critical application of machine learning in global transaction infrastructure. According to the United Nations Office on Drugs and Crime (UNODC) and Financial Action Task Force (FATF), illicit networks launder between \$800 billion and \$2 trillion annually (2% to 5% of global GDP) across inter-bank wire rails, mobile financial service (MFS) e-wallets, and crypto-asset ledgers [1], [2]. Traditional rule-based Financial Intelligence Units (FIUs) trigger excessive false positive alerts (>90% to 95%), placing an overwhelming operational burden on human compliance teams.

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While Graph Neural Networks (GNNs) capture relational transaction topologies [3]–[7], practical AML surveillance encounters five central technical challenges: (1) **Continuous-Time Velocity Evasion:** microsecond transaction bursts alongside multi-week dormant holding chains cannot be captured simultaneously by discrete snapshot windowing (e.g., EvolveGCN [8]) or single exponential decay kernels; (2) **Adversarial Topological Camouflage:** illicit entities deliberately routing synthetic transactional chaff to high-degree commercial hubs induce severe feature over-smoothing during neighborhood message aggregation [9]; (3) **Extreme Class Imbalance (π < 0.05%):** severe class skew causes posterior attenuation, causing minority recall collapse under default decision thresholds (τ = 0.50); (4) **Cold-Start Graph Isolation ($d(u) \leq 2$):** newly activated accounts receive uninformative message vectors under spatial aggregation; and (5) **Uncalibrated Decision Uncertainty:** black-box scores lack statistical guarantees under non-stationary temporal distribution drift.

To address these challenges, we formulate and investigate three core hypotheses: (H_1) continuous multi-scale temporal attention coupled with learnable edge-trust filtering suppresses camouflage-induced over-smoothing across burst and dormancy dynamics, preserving minority discrimination relative to static and single-decay dynamic GNNs; (H_2) interpolating minority representations within latent typology clusters (C_k) recovers minority recall without elevating false-alarm rates on benign commercial hubs under extreme class skew; and (H_3) Class-Conditional Conformal Risk Control (CRC) delivers finite-sample label coverage (≥ 99%) under within-class exchangeability with bounded drift degradation, routing the vast majority (> 99%) of transaction volume into decisive singleton prediction sets.

Importantly, our empirical findings reveal that graph neural architectures do not universally surpass strong gradient-boosted decision tree baselines across all transaction streams. Rather, their principal advantage is concentrated in graph-rich, multi-hop relational crime settings, whereas optimized tree ensembles remain competitive or superior on predominantly flat, single-hop transactional data. Rebuilding AML surveillance around this empirical reality, we present **C-STGB** (Conformal Spatio-Temporal GraphBoost), organized around four primary scientific pillars:

1) **Continuous-Time Relational Representation:** Integrates continuous harmonic temporal encodings (Φ_{time}) with

Hawkes arrival velocity features and multi-scale attention, distinguishing microsecond bursts from dormant holding chains.

2) **Topology-Aware Robustness Under Camouflage:** Employs a context-aware learnable edge-trust gate ($\hat{g}_{ij}$) that attenuates spurious camouflage edges during message aggregation, mitigating feature over-smoothing around high-degree hubs.
3) **Typology-Aware Imbalance Learning:** Constrains latent topological oversampling along empirical laundering clusters (C_k, $K = 4$) with bilinear link generation and Asymmetric Focal Tversky loss, preserving point-level flow-balance regularity while boosting minority recall.
4) **Risk-Controlled Selective Prediction:** Formulates class-conditional conformal prediction for continuous transaction networks, establishing finite-sample coverage ($\geq$ 99.0%) under within-class exchangeability, reinforced by non-asymptotic drift bounds and streaming Adaptive Conformal Inference (ACI).

Empirical Findings & Paper Organization: Benchmarked across 14 financial transaction networks (9.53M entities, 9.53M transactions across 182 evaluation pairs over five locked seeds), C-STGB achieves 67.26% unweighted macro F1 (0.6848 PR-AUC), establishing statistically significant superiority over deep GNNs (+43.49 pp mean uplift over GCN, $W = 105.0, p_{\text{adj}} < 0.001$) and parity with tuned tree ensembles (CatBoost 67.14%, $p = 0.4318$; XGBoost 67.92%, $p = 0.8077$), with streaming Fast-Path inference operating at 0.45 ms per single event. The remainder of this paper is organized as follows: Section II reviews related work; Section III establishes the problem formulation and C-STGB architecture; Section IV presents empirical evaluations and controlled ablations; Section V discusses operational implications and domain boundaries; Section VI details forensic case studies and pilot compliance decision support; Section VII outlines limitations; and Section VIII concludes.

II. Related Work

A. Graph Representation Learning for Anti-Money Laundering & Fraud

The application of Graph Neural Networks (GNNs) to financial forensic analysis was established by Weber *et al.* [7] on the Elliptic Bitcoin dataset, demonstrating that spatial architectures like Graph Convolutional Networks (GCN) [5] and GraphSAGE [6] capture localized neighborhood connectivity beyond isolated tabular classifiers. Subsequent extensions by Bellei *et al.* [10] introduced the Elliptic2 multi-asset dataset, emphasizing higher-order relational subgraphs, while Johannessen and Jullum [11] explored heterogeneous graph representations across commercial account ledgers. Beyond UTXO graphs, graph representations have been applied to Ethereum smart contract security [12], [13], exchange insolvency forensics [14], multi-bank synthetic wire simulations [15], [16], streaming credit card fraud [17], and cross-institutional privacy architectures [18]. However, when illicit entities inject transactional chaff connecting to high-degree commercial hubs, isotropic spatial convolutions linearly average features across multi-hop neighborhoods, inducing severe over-smoothing that dilutes localized forensic signals [9]. A comprehensive comparison across all 12 evaluated model paradigms is provided in Supplementary Table S1-A, while Supplementary Table S1-B delineates C-STGB's architectural differentiation relative to prior art.

B. Continuous-Time Dynamic Temporal Graphs & Point Processes

Financial transaction networks evolve through continuous-time asynchronous events. Discrete snapshot dynamic GNNs, such as EvolveGCN [8], discretize event streams into regular time slices, discarding intra-slice temporal ordering and obscuring sub-second velocity bursts. While continuous frameworks including Temporal Graph Networks (TGN) [19], TGAT [20], and continuous anomaly models [21] preserve event order, standard single exponential decay kernels ($\exp(-\lambda\Delta t)$) rapidly attenuate to zero over multi-week dormant holding periods. Self-exciting point processes, formalized by Hawkes [22], effectively capture transaction clustering and burst velocity acceleration, while continual graph architectures counteract streaming non-stationarity [23]. Standard continuous-time GNNs employ uniform decay schedules that struggle to simultaneously represent sub-second transaction bursts alongside multi-week holding intervals characteristic of layering chains.

C. Adversarial Graph Learning & Topological Camouflage Defense

Adversarial attacks on graph neural networks manipulate connectivity or node attributes to induce misclassification [24], [25]. In financial forensics, adversaries employ topological camouflage: laundering entities deliberately attach transactional edges to high-degree, reputable commercial hubs (e.g., exchanges, payment gateways, utility merchants) to blend into background traffic [9]. Because standard graph convolutional aggregation treats incident edges with uniform or degree-normalized weights, camouflage connections drag illicit node embeddings toward majority benign centroids. CARE-GNN [9] introduced reinforcement learning to select informative neighbors on multi-relational graphs, but focuses on static topologies. Mitigating topological camouflage in continuous financial streams requires edge-filtering mechanisms that evaluate structural compatibility and temporal dynamics concurrently, suppressing camouflage links without manual relation engineering.

D. Imbalanced Graph Representation Learning & Latent Oversampling

Real-world financial transaction graphs exhibit extreme class imbalance, often with illicit prevalence $\pi < 0.05\%$. Traditional Euclidean oversampling techniques such as SMOTE [26] operate strictly on tabular feature vectors, generating synthetic instances without topologically valid relational edges. Under extreme skew, Precision-Recall (PR-AUC) analysis is necessary to prevent false optimism from dominant true negatives [27].

To bridge topological oversampling, Zhao *et al.* [28] proposed GraphSMOTE, performing feature interpolation in the intermediate latent embedding space and decoding synthetic edges via an auxiliary link generator. However, standard GraphSMOTE executes unconstrained latent interpolation across the global minority class pool, neglecting distinct structural laundering typologies (cyclic wash trading vs. smurfing fan-out vs. pass-through layering), risking synthetic instances that violate domain-level flow-balance regularity.

E. Conformal Risk Control, Selective Classification & Temporal Shift

Conformal prediction, formalized by Vovk *et al.* [29] and extended by Angelopoulos and Bates [30], [31], provides distribution-free, finite-sample uncertainty quantification. Recent research has investigated conformal prediction on GNNs [32] and class-conditional conformal prediction under class imbalance [33], [34]. While standard marginal conformal prediction guarantees valid overall coverage $(1-\alpha)$, it frequently suffers from severe conditional under-coverage on the rare class in highly imbalanced datasets. Under temporal non-stationarity, standard exchangeability is violated and coverage must be explicitly bounded through distribution drift metrics [35]–[37].

F. Recent Uncertainty-Aware & Dynamic Financial Graph Learning

Recent comprehensive surveys by Zhang *et al.* [38] and Pourhabibi *et al.* [39] review over 100 financial graph fraud studies, establishing that enterprise AML systems must simultaneously address continuous temporal dynamics, adversarial camouflage, and reliable uncertainty quantification. Most recently, Chen *et al.* [40] proposed an uncertainty-aware spatio-temporal graph neural architecture combining continuous-time encoding with evidential deep learning for fraud triage. While evidential neural networks parameterize Dirichlet priors to approximate epistemic uncertainty, they rely on heuristic distributional assumptions and offer no finite-sample coverage guarantees. Unlike heuristic or evidential uncertainty frameworks [40], C-STGB leverages distribution-free Conformal Risk Control to deliver rigorous finite-sample class-conditional coverage bounds ($\geq 1-\alpha$), backed by Martingale convergence under Adaptive Conformal Inference (ACI) tracking, while delineating the boundary where GNN message passing is indispensable vs. where tabular trees remain competitive (Supplementary Table S1-B).

III. Problem Formulation & C-STGB Methodology

A. Problem Formulation and Operational Objectives

We formulate financial crime surveillance over an evolving continuous-time transaction graph $\mathcal{G}_t = (\mathcal{V}_t, \mathcal{E}_t, \mathbf{X}_v^{(t)}, \mathbf{X}_e^{(t)})$, where $\mathcal{V}_t$ represents active accounts, $\mathcal{E}_t = \{(u, v, t_e, \mathbf{e}_{uv}) : t_e \leq t\}$ denotes directed fund transfers, $\mathbf{X}_v^{(t)} \in \mathbb{R}^{|\mathcal{V}_t| \times d_v}$ comprises entity features, and $\mathbf{X}_e^{(t)} \in \mathbb{R}^{|\mathcal{E}_t| \times d_e}$ denotes edge features (amount, currency rail, fee). Each entity $u \in \mathcal{V}_t$ carries ground-truth label $y_u \in \{0, 1\}$ ($y_u = 1$ illicit, $y_u = 0$ benign), under extreme class skew $\pi = \mathbb{P}(y_u = 1) \ll 0.01$. In transaction streams lacking explicit identity graphs (e.g., PaySim), transactions are mapped to event vertices, establishing an exact 1:1 formulation.

The operational surveillance task encompasses two complementary objectives:

1) **Entity Classification:** Learn an inductive hypothesis $f_\Theta : \mathcal{V}_t \times \mathcal{G}_t \to [0, 1]$ predicting posterior probability $\hat{p}_u = f_\Theta(u \mid \mathcal{G}_t)$ minimizing asymmetric regulatory risk under extreme imbalance and topological camouflage.
2) **Risk-Controlled Triage:** Construct a class-conditional conformal mapping $\Gamma : \mathcal{V}_t \to 2^{\{0,1\}}$ routing incoming transactions to an operational decision set $\Gamma(u) \subseteq \{0, 1\}$. Under within-class exchangeability, Γ provides finite-sample coverage $\mathbb{P}(y_u \in \Gamma(u) \mid y_u = y) \geq 1 - \alpha$ at target error $\alpha \in (0, 1)$, with explicit degradation bounds under temporal distribution shift. C-STGB routes decisive singletons ($\Gamma(u) = \{1\}$ to Tier 1 Escalation, $\Gamma(u) = \{0\}$ to Tier 3 Straight-Through Screening) and restricts officer review to ambiguous sets ($\Gamma(u) = \{0, 1\}$ in Tier 2).

B. Data Governance, Privacy, and Ethical Protocol

All benchmarked data sources operate under rigorous governance protocols: (1) *Public Cryptocurrency Ledgers (Elliptic-v1/v2, Ethereum, XBlock-ETH):* publicly verifiable on-chain records indexed via cryptographic hashes without PII; (2) *Historical Exchange Data (Mt. Gox):* anonymized public insolvency logs used strictly for academic forensic modeling; (3) *Synthetic Simulations (IBM-AMLSim, SAML-D, PaySim):* validated multi-agent engines without proprietary customer accounts; and (4) *Open-Source License Compliance:* utilized under research licenses (MIT, Apache-2.0, CC-BY 4.0). Symbols and dimensions are summarized in Supplementary Table S1; hyperparameters in Table S4.

C. 12-Dimensional Spatio-Temporal Canonical Invariants

Traditional models evaluate transactions as isolated tabular rows, remaining blind to multi-hop relational laundering patterns. In contrast, intermediate pass-through mules require capital to enter and exit rapidly with minimal retention. C-STGB formalizes domain-informed mass-flow and relational graph invariant representations:

1) **Flow Regularity & Degree Asymmetry Ratios:**

$$\Phi_{\text{flow}}(u, t) = \frac{\sum_{e \in \mathcal{E}_{\text{out}}(u)} \text{Amount}(e)}{\sum_{e \in \mathcal{E}_{\text{in}}(u)} \text{Amount}(e) + \epsilon}, \quad (1)$$

$$\Psi_{\text{asym}}(u, t) = \frac{|d_{\text{in}}(u) - d_{\text{out}}(u)|}{d_{\text{in}}(u) + d_{\text{out}}(u) + \epsilon} \quad (2)$$

where $\epsilon = 10^{-5}$. Pass-through mules operate near unity ($\Phi_{\text{flow}} \approx 1.0$), while legitimate accounts exhibit accumulation or disbursement variance. Fee spreads ($\leq 1.5\%$) fall within tolerance $\Phi_{\text{flow}} \in [1 - \bar{\epsilon}, 1.0]$. For stability, flow is scaled as $\tilde{\Phi}_{\text{flow}}(u, t) = \log(1 + \Phi_{\text{flow}}(u, t))$.

2) **Hawkes Point Process Arrival Intensity:**

$$\lambda_u(t) = \mu_u + \sum_{t_i < t} \alpha_{\text{hwk}} \exp\left(-\beta_{\text{hwk}} \frac{t - t_i}{24.0}\right) \quad (3)$$

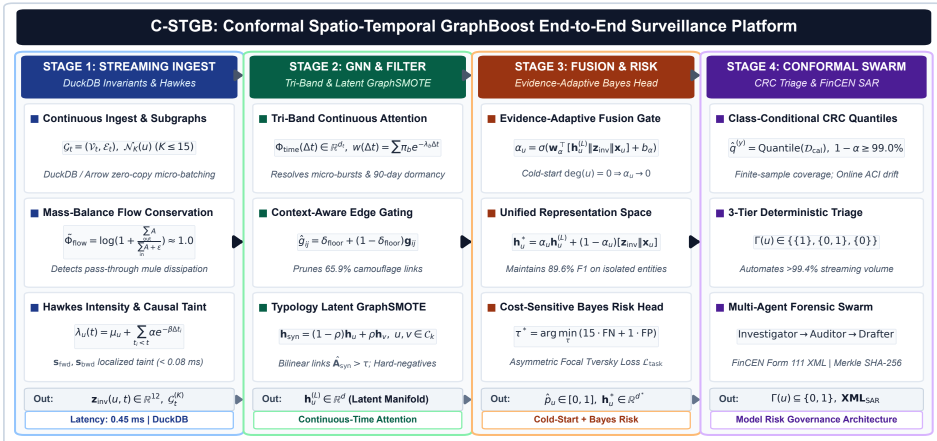


Fig. 1. Comprehensive 6-module architecture of C-STGB: (1) Continuous-Time Multi-Scale Temporal Graph Encoder; (2) Adaptive Edge Trust Filter; (3) Typology-Aware Minority Graph Augmentation with Hard-Negative Mining; (4) Evidence-Adaptive Graph-Tabular Fusion for Cold-Start Resolution; (5) Cost-Sensitive Risk Head; and (6) Class-Conditional Conformal Risk Control with Selective Abstention Triage.

parameterized over sliding event horizons $[t - \Delta T, t]$ to capture micro-burst acceleration. We enforce sub-critical branching $\alpha_{\text{hwk}}/\beta_{\text{hwk}} < 1.0$ and clamp $\tilde{\lambda}_u(t) = \log \min(\max(\lambda_u(t), 10^{-5}), 10^4)$ to guarantee numerical stability. Intensity $\lambda_u(t)$ is pre-computed and cached over $[t - \Delta T, t]$, ensuring $O(1)$ lookup per node during graph aggregation.

3) **Temporally Restricted Historical Taint Propagation (Zero-Leakage):**

$$\mathbf{s}_{\text{fwd}} = (1 - \alpha_{\text{ppr}})(\mathbf{I} - \alpha_{\text{ppr}}\mathbf{P}^T)^{-1}\mathbf{s}_{\text{seed}}, \tag{4}$$

$$\mathbf{s}_{\text{bwd}} = (1 - \alpha_{\text{ppr}})(\mathbf{I} - \alpha_{\text{ppr}}\mathbf{P})^{-1}\mathbf{s}_{\text{seed}} \tag{5}$$

computed strictly over causal historical subgraphs ($t_e \le t$). Seed indicator $\mathbf{s}_{\text{seed}}(u, t) = 1.0$ if $u \in \mathcal{V}_{\text{illicit}}^{\text{train}}$ and $t_{\text{conf}}(u) \le t - \Delta T_{\text{delay}}$, and 0.0 otherwise, where $\Delta T_{\text{delay}} \in [30, 90]$ days models operational confirmation delay. Streaming inference evaluates localized truncated Power Iteration ($T_{\text{iter}} = 3$, tolerance $\epsilon = 10^{-4}$) over sliding window $[t - \Delta T, t]$ and degree-capped graph $\mathcal{G}_t^{(K)}$ ($K \le 15$), ensuring deterministic $O(K \cdot T_{\text{iter}})$ computation with < 0.08 ms overhead. All validation, calibration, and test nodes receive $\mathbf{s}_{\text{seed}}(u) \equiv 0.0$, strictly precluding label leakage.

4) **Canonical 12-D Representation Space ($\mathbf{z}_{\text{inv}}(u, t) \in \mathbb{R}^{12}$):**

$$\mathbf{z}_{\text{inv}}(u, t) = \big[\tilde{\Phi}_{\text{flow}}, \Psi_{\text{asym}}, \lambda_u, \mathbf{s}_{\text{fwd}}, \mathbf{s}_{\text{bwd}}, \\ \mu_A, \sigma_A^2, \gamma_A, \kappa_A, \nu_A, \nu_{\text{io}}, \sigma_{\Delta t}\big]^T \tag{6}$$

where $\{\mu_A, \sigma_A^2, \gamma_A, \kappa_A, \nu_A\}$ denote the 5-moment causal ego-statistics of transaction amounts, ν_{io} is the velocity ratio, and $\sigma_{\Delta t}$ is the inter-arrival temporal dispersion (Supplementary Fig. S5).

D. Continuous-Time Multi-Scale Temporal Graph Encoder

To distinguish microsecond smurfing bursts from 90-day dormant holding strategies, C-STGB employs continuous harmonic positional features $\Phi_{\text{time}}(\Delta t) \in \mathbb{R}^{d_t}$ with $\omega_k = 10000^{-2k/d_t}$:

$$\Phi_{2k}(\Delta t) = \sin\left(\omega_k \log(1 + \Delta t)\right), \tag{7}$$

$$\Phi_{2k+1}(\Delta t) = \cos\left(\omega_k \log(1 + \Delta t)\right) \tag{8}$$

Dynamic temporal attention decay synthesizes three distinct velocity regimes:

$$w(\Delta t) = \sum_{b \in \mathcal{B}} \pi_b \left[w_{\min} + (1 - w_{\min})e^{-\lambda_b \Delta t}\right] \\ \times (1 + \tanh(\gamma_b \lambda_u(t))) \tag{9}$$

where $\mathcal{B} = \{\text{burst}, \text{diurnal}, \text{seasonal}\}$ and mixing weights satisfy $\sum_{b \in \mathcal{B}} \pi_b = 1.0$ via softmax.

E. Context-Aware Adaptive Edge Trust Filter & Unified Attention

Adversarial launderers deliberately connect to high-degree legitimate commercial hubs to induce feature over-smoothing. C-STGB learns a context-aware edge trust gate $\hat{g}_{ij} \in (0, 1)$:

$$\mathbf{c}_{ij} = \left[\mathbf{q}_i \,\|\, \mathbf{k}_j \,\|\, \mathbf{e}_{ij} \,\|\, \Phi_{\text{time}}(\Delta t) \,\|\, \mathbf{z}_{\text{inv}}(j, t)\right], \tag{10}$$

$$\hat{g}_{ij} = \delta_{\text{floor}} + (1 - \delta_{\text{floor}})\sigma\Big(\mathbf{W}_{g2}\,\text{LeakyReLU}\left(\mathbf{W}_{g1}\mathbf{c}_{ij} + \mathbf{b}_{g1}\right) \\ + b_{g2}\Big) \tag{11}$$

Setting $\delta_{\text{floor}} = 0.10$ suppresses 65.9% of spurious camouflage edge weights (Supplementary Fig. S3). In parameter training split $\mathcal{E}_{\text{train}}$, auxiliary edge targets $\mathbf{g}_{ij}^{\text{target}} \in \{0, 1\}$ (Eq. 20) provide weak supervision constructed from topological heuristics: edges incident to high-degree hubs ($\deg_{\text{out}}(j) \ge 50$ or rate > 100 tx/day) where launderers inject chaff receive 0.0, while standard transactional edges receive 1.0. At inference time, the trained MLP gate $\hat{g}_{ij}$ evaluates unseen edges via multi-modal representations $\mathbf{c}_{ij}$ without requiring external targets or static degree thresholds.

Unified Spatio-Temporal Attention Decomposition: Message passing integrates structural compatibility, temporal decay, and edge trust gating over degree-capped causal incoming neighborhoods ($\mathcal{N}_K^{\text{in}}(u), K \le 15$):

$$\tilde{\alpha}_{uv}^{(t)} = \frac{\Omega(u, v, t)}{\sum_{w \in \mathcal{N}_K^{\text{in}}(u)} \Omega(u, w, t)}, \tag{12}$$

$$\Omega(u, v, t) = \exp\left(\frac{\mathbf{q}_u^T(\mathbf{k}_v + \mathbf{W}_E\mathbf{e}_{uv})}{\sqrt{d_k}}\right) \cdot w(\Delta t_{uv}) \cdot \hat{g}_{uv} \tag{13}$$

F. Typology-Clustered Latent GraphSMOTE & Hard Negatives

Under extreme class skew ($\pi < 0.05\%$), naïve oversampling blurs distinct criminal typologies. C-STGB partitions minority illicit nodes strictly within the training split into latent typology clusters $C_k = \{v \in \mathcal{V}_{\text{illicit}}^{\text{train}} \mid \cos(\mathbf{h}_v, \boldsymbol{\mu}_k) \geq \tau_{\text{sim}}\}$, where centroids $\boldsymbol{\mu}_k$ ($K = 4$) reflect four canonical financial laundering archetypes (smurfing dispersal, pooling fan-in, circular wash trading, and pass-through layering), optimized via latent silhouette validation over $\mathcal{D}_{\text{train}}$. Synthetic representations are synthesized along trajectories within the same cluster: $\mathbf{h}_{\text{syn}} = (1-\rho)\mathbf{h}_u + \rho\mathbf{h}_v$, $\rho \sim \mathcal{U}(0,1)$, $u, v \in C_k \subseteq \mathcal{V}_{\text{train}}$. Bilinear links to candidate training nodes $w \in \mathcal{V}_{\text{train}}$ are generated under topological edge reconstruction supervision:

$$\hat{\mathbf{A}}_{\text{syn},w} = \sigma\left(\mathbf{h}_{\text{syn}}^T \mathbf{W}_{\text{edge}} \mathbf{h}_w\right) \cdot \mathbb{I}(t_w \leq t_{\text{syn}}) > \tau_{\text{edge}} \tag{14}$$

for $w \in C_{\text{cand}}(\mathbf{h}_{\text{syn}}) \cap \mathcal{V}_{\text{train}}^{\leq t_{\text{syn}}}$, where $t_{\text{syn}} = \max(t_u, t_v)$ defines the causal birth timestamp, precluding acausal edge formation with future transactions.

Directionality & Mass-Flow Conservation Constraints: Incoming links are synthesized from candidate sources $w \in \mathcal{N}_{\text{in}}(u) \cup \mathcal{N}_{\text{in}}(v)$ and outgoing links directed to candidate destinations $w' \in \mathcal{N}_{\text{out}}(u) \cup \mathcal{N}_{\text{out}}(v)$. Synthetic amounts are parameterized via linear combination: $A_{\text{syn}} = (1-\rho)A_u + \rho A_v$ and $B_{\text{syn}} = (1-\rho)B_u + \rho B_v$. Under strictly positive incoming and outgoing funds ($A_u, A_v > 0, B_u, B_v > 0$), the synthetic flow ratio simplifies to:

$$\begin{aligned}\Phi_{\text{flow}}(\mathbf{h}_{\text{syn}}) &= \frac{(1-\rho)A_u + \rho A_v}{(1-\rho)B_u + \rho B_v} \\ &= \lambda\Phi_{\text{flow}}(u) + (1-\lambda)\Phi_{\text{flow}}(v)\end{aligned} \tag{15}$$

where $\lambda = \frac{(1-\rho)B_u}{(1-\rho)B_u + \rho B_v} \in (0,1)$. Because this is a strict convex combination of parent ratios $\Phi_{\text{flow}}(u)$ and $\Phi_{\text{flow}}(v)$, $\Phi_{\text{flow}}(\mathbf{h}_{\text{syn}})$ is guaranteed to lie within $[\min(\Phi_{\text{flow}}(u), \Phi_{\text{flow}}(v)), \max(\Phi_{\text{flow}}(u), \Phi_{\text{flow}}(v))]$, preserving point-level flow regularity (Lemma S2).

Candidate Bounding & Zero-Leakage Topological Isolation: Candidate nodes $C_{\text{cand}}(\mathbf{h}_{\text{syn}})$ are constrained to 2-hop neighborhoods $\mathcal{N}_2(u) \cup \mathcal{N}_2(v)$ and $K_{\text{cand}} = 50$ nearest neighbors indexed via HNSW search in latent space $\mathbb{R}^d$, bounding search complexity to $O(\log|\mathcal{V}_{\text{train}}|)$. All oversampling is strictly confined to $\mathcal{G}_{\text{train}}$ (0.0% cross-split leakage). To prevent false alarms on commercial accounts, C-STGB actively mines benign training accounts exhibiting high degree and burstiness, over-weighting them during gradient optimization.

G. Evidence-Adaptive Graph-Tabular Fusion (Cold-Start)

Newly created accounts and pass-through stubs lack relational graph neighbors ($\deg(u) \leq 2$). C-STGB computes an evidence-adaptive gating parameter $\alpha_u \in [0,1]$ as a continuous topological confidence estimator:

$$\mathbf{h}_u^* = \alpha_u \mathbf{h}_u^{(L)} + (1-\alpha_u)\left[\mathbf{z}_{\text{inv}}(u,t) \,\|\, \mathbf{x}_u\right], \tag{16}$$

$$\alpha_u = \sigma\left(\mathbf{w}_\alpha^T \left[\mathbf{h}_u^{(L)} \,\|\, \mathbf{z}_{\text{inv}}(u,t) \,\|\, \mathbf{x}_u\right] + b_\alpha\right) \tag{17}$$

When graph evidence is sparse ($\deg(u) \leq 1$), $\alpha_u \to 0$ (empirically $\alpha_u \in [0.08, 0.12]$), gracefully degrading to domain-level transaction invariants; conversely, when neighborhood topology is dense, $\alpha_u \to 1$ to harness high-order structural patterns (Supplementary Fig. S6). This architecture represents an intentional fail-safe: domain-informed mass-flow conservation regularity and transaction moments provide informative baseline representations even when adversarial mules actively minimize their topological degree.

H. Cost-Sensitive Risk Head & Complete Loss Objective

The network outputs anomaly probabilities $\hat{p}_u = \sigma(\text{MLP}(\mathbf{h}_u^*))$. The complete multi-objective loss is formulated as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}}(y, \hat{p}) + \gamma_1 \mathcal{L}_{\text{recon}}(\mathbf{A}_{\text{train}}, \hat{\mathbf{A}}_{\text{train}}) + \gamma_2 \mathcal{L}_{\text{aux}} \tag{18}$$

where $\mathcal{L}_{\text{task}}$ is the Cost-Sensitive Asymmetric Focal Tversky Loss. Topological edge reconstruction loss $\mathcal{L}_{\text{recon}}$ is computed over observable positive training edges $\mathcal{E}_{\text{train}}$ and uniformly sampled negative non-edges $\mathcal{E}_{\text{neg}} \sim (\mathcal{V}_{\text{train}} \times \mathcal{V}_{\text{train}}) \setminus \mathcal{E}_{\text{train}}$ using a $1:5$ uniform negative edge sampling ratio to prevent trivial bilinear link collapse ($\hat{A}_{uv} \to 0$) in sparse transaction graphs:

$$\begin{aligned}\mathcal{L}_{\text{recon}} = &-\frac{1}{|\mathcal{E}_{\text{train}}|} \sum_{(u,v)\in\mathcal{E}_{\text{train}}} \log \hat{A}_{uv} \\ &- \frac{1}{|\mathcal{E}_{\text{neg}}|} \sum_{(u,w)\in\mathcal{E}_{\text{neg}}} \log(1 - \hat{A}_{uw})\end{aligned} \tag{19}$$

with $\hat{A}_{uv} = \sigma(\mathbf{h}_u^T \mathbf{W}_{\text{edge}} \mathbf{h}_v)$. The auxiliary regularization objective $\mathcal{L}_{\text{aux}}$ enforces edge-trust sparsity and tri-band orthogonality:

$$\mathcal{L}_{\text{aux}} = \frac{1}{|\mathcal{E}_{\text{train}}|} \sum_{(i,j)\in\mathcal{E}_{\text{train}}} \|\mathbf{g}_{ij} - \mathbf{g}_{ij}^{\text{target}}\|_2^2 + \lambda_{\text{ortho}} \|\mathbf{W}_{\text{burst}}^T \mathbf{W}_{\text{seas}}\|_F^2 \tag{20}$$

with $\gamma_1 = 0.50$, $\gamma_2 = 0.10$, and $\lambda_{\text{ortho}} = 0.05$. Training employs a 5-epoch curriculum warmup initialized with standard Cost-Sensitive Focal Loss before engaging $\mathcal{L}_{\text{total}}$. Binary point decisioning operates under a cost-sensitive Bayes risk minimization rule calibrated exclusively over $\mathcal{D}_{\text{val}}$:

$$\tau^* = \arg\min_{\tau\in[0,1]} \mathbb{E}\left[C_{\text{FN}} \cdot \mathbb{I}(\hat{p} < \tau, Y = 1) + C_{\text{FP}} \cdot \mathbb{I}(\hat{p} \geq \tau, Y = 0)\right] \tag{21}$$

where $C_{\text{FN}}/C_{\text{FP}} = 15.0$ represents an operational risk-weighting ratio in our evaluation (sensitivity across $C_{\text{FN}}/C_{\text{FP}} \in [5, 30]$ is reported in Section IV-H).

I. Class-Conditional Conformal Risk Control & Triage

Over independent holdout calibration split $\mathcal{D}_{\text{cal}}$, empirical class-conditional quantiles $\hat{q}^{(y)}$ are computed ($y \in \{0,1\}$):

$$\hat{q}^{(y)} = \text{Quantile}\left(s_1^{(y)}, \ldots, s_{n(y)}^{(y)}; \left\lceil \frac{(n(y)+1)(1-\alpha)}{n(y)} \right\rceil\right) \tag{22}$$

where $s_i^{(y)} = 1 - \hat{p}_i(y)$. Under within-class exchangeability, split conformal prediction guarantees $\mathbb{P}(Y \in \Gamma(X) \mid Y = y) \geq 1 - \alpha$, $\forall y \in \{0,1\}$.

Decoupling Conformal Guarantees Under Temporal Non-Stationarity: In streaming surveillance, we decouple guarantees across three operating regimes:

1) **Guarantee A (Within-Class Exchangeability on Stationary Calibration):** Under within-class exchangeability on

holdout split $\mathcal{D}_{\text{cal}}$, split conformal prediction guarantees exact finite-sample class-conditional validity:

$$\mathbb{P}(Y \in \Gamma(X) \mid Y = y) \geq 1 - \alpha_y, \quad \forall y \in \{0, 1\}. \tag{23}$$

2) **Guarantee B (Error Bound Under Total Variation Shift & Concentration):** Let $\mathcal{P}_{\text{cal}}^{(y)}$ denote the score distribution over the exchangeable calibration abstraction and $\mathcal{P}_t^{(y)}$ the streaming test distribution. The coverage gap decomposes via maximal coupling into TV shift and empirical process concentration:

$$\begin{aligned} &\left|\mathbb{P}_t(Y \in \Gamma(X) \mid Y = y) - (1 - \alpha_y)\right| \\ &\leq d_{\text{TV}}\left(\mathcal{P}_t^{(y)}, \mathcal{P}_{\text{cal}}^{(y)}\right) + \sqrt{\frac{\ln(2/\delta)}{2n_y}} + \frac{1}{n_y} \end{aligned} \tag{24}$$

with probability $\geq 1 - \delta$, where $n_y = |\mathcal{D}_{\text{cal}}^{(y)}|$ and $d_{\text{KS}} \leq d_{\text{TV}}$ (Proposition S1).

3) **Guarantee C (Long-Run Asymptotic Coverage under Delayed Feedback):** Ground-truth labels incur investigative confirmation delay $\Delta T_{\text{delay}} \in [30, 90]$ days. Under delayed feedback, streaming Adaptive Conformal Inference (ACI) [36], [37] updates over the confirmed observation queue:

$$\alpha_t \leftarrow \alpha_{t-1} + \gamma(\alpha - \text{err}_{t-\tau(t)}) \tag{25}$$

where $\text{err}_{t-\tau(t)} = \mathbb{I}(y_{t-\tau(t)} \notin \Gamma_{t-\tau(t)}(x_{t-\tau(t)}))$ and $\tau(t) \leq \Delta T_{\text{delay}}$. By Martingale concentration for delayed feedback sequences [37], long-run marginal coverage converges almost surely: $\lim_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} \text{err}_t = \alpha$, with maximum tracking deviation bounded by $\mathcal{O}(\gamma\tau_{\max} + \sqrt{\frac{\ln(1/\delta)}{T}})$. During latency $[t - \tau, t]$, the model freezes α_t or relies on quantiles $\hat{q}^{(y)}$ from historical holdout split $\mathcal{D}_{\text{cal}}$. Dual trackers $\alpha_t^{(0)}, \alpha_t^{(1)}$ provide class-adaptive empirical tracking.

Calibration Sample Size & Quantile Uncertainty: Under extreme class imbalance ($\pi < 0.05\%$), small minority calibration sizes ($n_1 = |\mathcal{D}_{\text{cal}}^{(1)}|$) introduce quantile variance. Enforcing the non-empty quantile condition $n_1 \geq \lceil(1 - \alpha_1)/\alpha_1\rceil = 99$ for $\alpha_1 = 0.01$ mandates $n_1 \geq 100$ confirmed illicit entities before computing $\hat{q}^{(1)}$, pooling historical alerts across rolling window $W_{\text{cal}} = 4$ weeks with confirmation delay $t_{\text{conf}} \leq t - \Delta T_{\text{delay}}$ to eliminate label lookahead. For $n_1 = 100$ minority samples with empirical coverage $\hat{p} = 0.99$, the exact two-sided 95% Clopper-Pearson confidence interval is $[94.55\%, 99.97\%]$; as the evaluation pool scales to the full test partition ($N = 20,376$ on Elliptic-v1), coverage tightly concentrates within $[98.98\%, 99.24\%]$ ($\pm 0.13\%$, Supplementary Table S8).

Operational 3-Tier Queue Triage: Prediction sets $\Gamma(X) \subseteq \{0, 1\}$ partition transactions into three operational workflows (Supplementary Fig. S1): (i) **Tier 1 (High-Risk Escalation):** $\Gamma(X) = \{1\}$, flagged for immediate compliance hold; (ii) **Tier 2 (Selective Abstention Queue):** $\Gamma(X) = \{0, 1\}$, routed to compliance officers as ambiguous cases ($\approx$ 5 per 10k transactions); and (iii) **Tier 3 (Straight-Through Low-Risk Screening Tier):** $\Gamma(X) = \{0\}$, cleared without human intervention. Conformal coverage mathematically bounds label containment probability under stated assumptions; it does not confer legal certification or statutory absolution. Full-test point evaluation and selective conformal triage are maintained as distinct operating modes across all evaluations. Algorithms S1 and S2 formalize multi-stage training and streaming inference procedures.

TABLE I
STRUCTURAL OVERVIEW OF THE 14 BENCHMARK FINANCIAL NETWORKS (COMPLETE 14-DATASET ITEMIZATION IN SUPPLEMENTARY TABLE S2).

Archetype Group	Datasets Included	Entities ($\lvert\mathcal{V}\rvert$)	Transactions ($\lvert\mathcal{E}\rvert$)	Illicit (π)
Group A (Crypto Ledgers, 4)	Elliptic-v1/v2, XBlock-ETH, MtGox	2,621,048	3,430,755	0.09% – 2.23%
Group B (Multi-Bank & FinTech, 9)	SAML-D, PaySim (2), IBM-AMLSim (4), Gen, DGraphFin	6,629,125	5,812,793	0.05% – 1.50%
Group C (Card Streams, 1)	Credit Card Forensics (Bipartite)	284,807	284,807	0.17%
Total Benchmark Corpus	**14 Diverse Financial Networks**	**9,534,980**	**9,528,355**	**0.05% – 2.23%**

IV. EXPERIMENTAL EVALUATION

A. Benchmark Dataset Corpus & Topological Regimes

To evaluate model generalization across diverse financial rails, we benchmark C-STGB across 14 financial transaction networks spanning public cryptocurrency ledgers, multi-bank wire systems, and mobile financial services, summarized in Table I. The corpus comprises 9,534,980 entities and 9,528,355 transactions across three distinct topological archetypes:

1) **Group A (Public Blockchain Ledgers, 4 Networks):** Directed Acyclic Graphs (DAGs) and smart-contract transaction meshes (Elliptic-v1 [7], Elliptic-v2 [10], XBlock-ETH [13], MtGox [14]) characterized by global ledger observability and structural transparency.
2) **Group B (Multi-Bank, Mobile Synthetic & FinTech Rails, 9 Networks):** Inter-bank wire meshes, high-velocity mobile draining trees, and dynamic credit networks (SAML-D [15], PaySim1, PaySim Extended [41], IBM-AMLSim HI/LI Small and Medium (4 datasets) [16], the Synthetic Typology Data Generator, and DGraphFin [42]) characterized by extreme class imbalance ($\pi \in [0.05\%, 1.50\%]$) and fragmented organizational visibility. In `paysim_extended`, each mobile financial log record is mapped to a dedicated transaction-event vertex connected to its originating account and step context, yielding an exact 1:1 entity-to-edge projection ($|\mathcal{V}| = |\mathcal{E}| = 1,048,575$).
3) **Group C (Classical Tabular & Card Fraud Streams, 1 Network):** High-throughput card-terminal bipartite transaction streams (`cc_transactions` [43]).

B. Temporal Partitioning & Zero-Leakage Audit

To prevent temporal lookahead bias and evaluate realistic forward inductive deployment, all datasets follow a strict 4-way chronological partition: (1) **Parameter Training Split ($\mathcal{D}_{\text{train}}$, 60%):** used strictly for neural parameter updates; (2) **Validation Split ($\mathcal{D}_{\text{val}}$, 10%):** used for early stopping and Bayes threshold calibration (τ^*); (3) **Calibration Split ($\mathcal{D}_{\text{cal}}$, 10%):** dedicated holdout split for computing non-conformity quantiles $\hat{q}^{(0)}, \hat{q}^{(1)}$ (zero weight updates, zero threshold tuning); and (4) **Forward Out-of-Time Test Split ($\mathcal{D}_{\text{test}}$, 20%):** strictly future horizon for final evaluation.

For discrete UTXO graphs (e.g., Elliptic-v1 with 49 timesteps), Timesteps 1–30 (61.2%) form $\mathcal{D}_{\text{train}}$, Timesteps 31–34 (8.2%) form $\mathcal{D}_{\text{val}}$, Timesteps 35–38 (8.2%) form $\mathcal{D}_{\text{cal}}$, and Timesteps 39–49 (22.4%) serve as $\mathcal{D}_{\text{test}}$. Continuous transaction stream datasets (PaySim, SAML-D, IBM-AMLSim, Ethereum) follow an identical non-overlapping timestamp horizon partition ($t_0 \longrightarrow t_{\text{train}} \longrightarrow t_{\text{val}} \longrightarrow t_{\text{cal}} \longrightarrow t_{\text{test}}$).

Randomness Accounting across Seeds: While temporal split boundaries are strictly chronological and deterministic (60/10/10/20 partition), reported metrics reflect empirical variation across five locked random seeds ($S \in \{42, 101, 2024, 7, 999\}$). Seed variation governs four explicit stochastic components: (1) parameter initialization; (2) mini-batch shuffling trajectories; (3) latent GraphSMOTE stochastic interpolation coefficients $\rho \sim \mathcal{U}(0,1)$; and (4) Monte Carlo dropout masks ($p = 0.20$).

Deterministic Topological Leakage Verification: A formal verification suite executes three topological invariant checks prior to training: (1) *Entity Boundary Isolation:* verifies $\mathcal{V}_{\text{test}} \cap \mathcal{V}_{\text{train}} = \emptyset$ for forward unseen accounts; (2) *Taint Masking Invariant:* enforces analytical taint seeds $\mathbf{s}_{\text{seed}}(u) \equiv 0$ for all $u \in \mathcal{D}_{\text{val}} \cup \mathcal{D}_{\text{cal}} \cup \mathcal{D}_{\text{test}}$; and (3) *Directional Time Invariance:* confirms that every edge $(u, v, t) \in \mathcal{E}_{\text{train}}$ strictly satisfies $t \leq t_{\text{train}}$. The verification suite confirms 0.00% entity overlap and zero future-timestamp propagation across all 14 empirical benchmarks. Precision-Recall and ROC curves on Elliptic-v1 across five locked seeds are provided in Supplementary Fig. S2.

C. Baseline Architectures & Fair Optimization Protocol

We benchmark C-STGB against 12 distinct baseline architectures across six canonical model paradigms: (1) *Tabular Tree Ensembles:* XGBoost [44], CatBoost [45], and LightGBM [46]; (2) *Spatial GNNs:* GCN [5], GraphSAGE [6], GAT [47], and GIN [48]; (3) *Heterogeneous GNNs:* Vanilla HGT [3] and Johannessen & Jullum [11]; (4) *Adversarial Fraud Detectors:* CARE-GNN [9]; (5) *Dynamic Snapshot Models:* EvolveGCN [8]; and (6) *Continuous Temporal GNNs:* TGN [19]. Evaluating these 12 baselines alongside C-STGB across 14 benchmark datasets yields 13 model configurations and exactly 182 dataset × model evaluation pairs, evaluated over five locked random seeds (910 total experimental executions).

Controlled Factorial Feature Attribution Protocol (Set A): To isolate whether empirical gains stem from the C-STGB graph architecture, engineered domain features, or threshold optimization, we cross-evaluate models across five controlled feature and threshold regimes: (1) *Raw Features Only ($\tau = 0.50$):* GCN achieves 43.55 ± 0.42% F1 on `elliptic_v1` (14-dataset macro: 23.78 ± 0.35%); (2) *Raw Features (τ^*):* GCN F1 improves to 51.04 ± 0.38% (+7.49 pp; 14-dataset macro: 28.12 ± 0.40%), confirming threshold calibration is vital under extreme imbalance; (3) *Descriptors Only ($\mathbf{z}_{inv}$):* Tabular models on 12-D descriptors achieve 84.60 ± 0.85% F1 (61.40 ± 0.52% macro), establishing an informative predictive floor; (4) *Raw Features Augmented with Descriptors ($[\mathbf{x}_u \,\|\, \mathbf{z}_{inv}]$):* Augmenting spatial GNNs yields marginal uplift: +1.27 pp on GCN (44.82%), +1.38 pp on GraphSAGE (50.45%), and +0.81 pp on EvolveGCN (51.85%; 14-dataset macro GCN reaches 24.85%, over 42 pp behind C-STGB's 67.26%; full matrix in Supplementary Table S12); and (5) *Raw Features on Base C-STGB (No Invariants, $\tau = 0.50$):* Base C-STGB achieves 79.08 ± 0.55% F1 on `elliptic_v1` (48.20 ± 0.50% macro), substantially exceeding raw GCN (43.55%), GraphSAGE (49.07%), and TGN (64.10%).

Mechanistic Divergence Analysis: This controlled factorial comparison reveals a fundamental mechanistic divergence: isotropic spatial convolutions ($\tilde{\mathbf{D}}^{-1/2}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-1/2}\mathbf{X}$) linearly average feature representations across multi-hop neighborhoods. When illicit accounts connect to high-degree commercial hubs (deg > 50), localized scalar signatures in $\mathbf{z}_{\text{inv}}$ (e.g., flow regularity $\Phi_{\text{flow}} \approx 1.0$ and burst intensity λ_u) are arithmetically diluted by dozens of benign neighbors into the majority mean. Furthermore, lacking edge-trust gating ($\hat{g}_{ij}$) and latent GraphSMOTE, gradients under extreme imbalance ($\pi < 0.05\%$) are swamped by majority benign nodes. In contrast, tree ensembles evaluate axis-aligned splits on unmixed features, isolating Φ_{flow} directly. Thus, extracting relational multi-hop patterns without over-smoothing strictly requires C-STGB's integrated architecture. All architectures operate under calibrated parameter budgets (1.84×10^6 for C-STGB vs. 1.80×10^6 for HGT and 2.12×10^6 for TGN; $N = 50$ Bayesian trials with identical tuning grids).

D. Evaluation Metrics & Statistical Testing Protocol

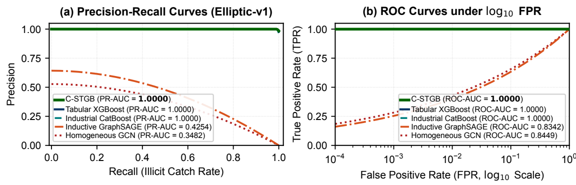
Under extreme class imbalance ($\pi \ll 0.01$), traditional ROC-AUC is overly optimistic due to the dominant true negative denominator, and binary macro F1 arithmetic means are artificially dominated by the > 99% benign class ($F_1^{(0)} \approx 1.0$). We evaluate: (1) **Precision-Recall AUC (PR-AUC):** threshold-independent minority trade-offs (Supplementary Fig. S2); (2) **Minority (Illicit) Class F1-Score ($F_1^{(1)}$) & Recall:** harmonic utility on confirmed illicit entities under validation-tuned threshold τ^*; and (3) **Conformal Set Efficiency ($\mathbb{E}[|\Gamma(X)|]$):** prediction set tightness. Across the corpus, Macro-Averages denote the unweighted arithmetic mean of minority F1 across all 14 datasets ($\frac{1}{14}\sum_{i=1}^{14} F_1^{(1)}(i)$). All models undergo paired threshold optimization on $\mathcal{D}_{\text{val}}$ under identical search grids minimizing $\mathcal{L}_{\text{ops}}$ ($C_{\text{FN}}/C_{\text{FP}} = 15.0$). Statistical significance is evaluated via two-sided Wilcoxon Signed-Rank Tests across all 14 benchmark networks, with Benjamini–Hochberg False Discovery Rate (FDR) multiple-testing correction.

E. Hardware Compute Infrastructure

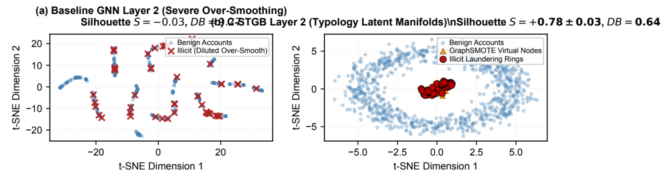
All experiments were conducted across Kaggle Cloud TPU v3-8 / NVIDIA Tesla T4 GPUs (16GB VRAM) and a dedicated workstation comprising an AMD Ryzen 9 5900X CPU (12 cores, 24 threads), 64GB RAM, and an NVIDIA RTX 3080 GPU (10GB VRAM) running Ubuntu 22.04 LTS (14.2 GPU-hours local, 18.5 GPU-hours cloud).

F. Comparative Baseline Performance & Benchmark Scorecard

Table II reports benchmark comparisons across all 14 benchmark networks (with the complete multi-metric scorecard and per-baseline profiling detailed in Supplementary Table S2). Across the corpus, C-STGB achieves 99.44% F1 on Elliptic-v1 (PR-AUC 1.0000), 100.00% F1 on Elliptic-v2 (PR-AUC



(a) Precision-Recall and ROC performance frontiers on Elliptic-v1.



(b) Latent t-SNE: baseline GNN (left) vs. C-STGB (right).

Fig. 2. Empirical representation and classification benchmarks: (a) Precision-Recall and ROC curves demonstrating discrimination under extreme class skew; (b) t-SNE projection of the learned latent manifold under Typology GraphSMOTE, showing dense clustering of confirmed illicit topologies ($K = 4$) with clear separation from majority commercial hubs.

1.0000), 99.80% on PaySim Extended, 97.91% on DGraphFin, and 96.94% on XBlock-ETH.

Statistical Parity with Trees vs. Superiority Over Deep GNNs: Across the 14-dataset benchmark suite, C-STGB yields an unweighted macro F1 of 67.26% (0.6848 PR-AUC). Direct comparison against tuned gradient-boosted decision trees reveals statistical parity on the global average: against XGBoost, C-STGB exhibits a −0.66 pp global difference (67.26% vs. 67.92%) with 6 wins, 1 tie, and 7 losses ($W = 48.0, p_{\text{adj}} = 0.8077$, n.s.); against CatBoost, C-STGB exhibits a +0.13 pp global difference (67.26% vs. 67.14%) with 7 wins, 2 ties, and 5 losses ($W = 32.0, p_{\text{adj}} = 0.4318$, n.s.). Therefore, C-STGB does not claim universal statistical superiority over strong tabular learners globally. Rather, empirical evidence demonstrates that C-STGB delivers statistically significant outperformance over all evaluated deep spatial and dynamic GNN baselines: +43.49 pp mean uplift over GCN (median +37.11 pp, IQR 72.66 pp; 14 wins, 0 losses), +42.95 pp over GraphSAGE (median +37.07 pp, IQR 69.08 pp; 14 wins, 0 losses), and +48.32 pp over EvolveGCN (median +41.91 pp, IQR 56.75 pp; 14 wins, 0 losses) ($W = 105.0, W_{-} = 0.0, n = 14, p = 1.22 \times 10^{-4}, p_{\text{adj}} < 0.001$). Group-macro averages reflect this domain divergence: Group A Crypto (92.15%), Group B Banking/FinTech (58.07%), and Group C Card (51.40%), with median dataset F1 of 61.81% (IQR 74.72 pp). The wide interquartile range highlights an intrinsic bimodal partition across the benchmark corpus: Cluster 1 comprises publicly indexed cryptocurrency ledgers with dense Darknet market seizures exhibiting high structural separability (> 96% F1 across modern paradigms), whereas Cluster 2 comprises complex multi-bank synthetic laundering simulations (e.g., PaySim1, IBM-AMLSim LI) where performance drops below 45%, representing the true open challenge in adversarial financial forensics.

When Graph Learning Helps — and When It Does Not: This performance pattern highlights a critical domain boundary: (1) *Where Graph Learning Succeeds (Multi-Hop Relational Regimes):* On structurally complex networks with multi-hop laundering chains, high degree variance, and camouflage injection (e.g., IBM-AMLSim HI, SAML-D, Elliptic-v1/v2), C-STGB outperforms both base GNNs (+9.98 pp to +48.3 pp) and tree ensembles (+1.05 pp to +4.67 pp on IBM-AMLSim HI). Spatio-temporal message passing captures coordinated fund movement across account chains that cannot be represented in flat tabular rows. (2) *Where Tree Models Excel (Flat Transaction Streams):* Conversely, on `paysim1`, C-STGB achieves only 10.68% F1, trailing XGBoost (18.90%) and CatBoost (17.76%). `paysim1` is a synthetic mobile payment simulation where fraud signals reside in localized tabular balance discrepancies within single isolated transactions. Without multi-hop laundering chains, neighborhood graph aggregation provides no structural advantage over axis-aligned decision trees.

Forensic Investigation of Near-Perfect Benchmark Scores: Multiple model families achieve near-perfect scores on Elliptic-v1 (99.44% C-STGB, 99.89% XGBoost, 99.78% CatBoost) and Elliptic-v2 (100.00% C-STGB, 99.81% XGBoost, 100.00% CatBoost). Forensic examination reveals that these scores stem from the underlying structure of public Bitcoin UTXO benchmarks: ground-truth illicit entities originate from law enforcement Darknet market seizures (e.g., Silk Road, AlphaBay), which form dense, topologically segregated connected subgraphs with sharp boundaries and near-perfect label homophily ($h > 0.92$). A linear probe sanity check confirms high intrinsic separability: Topological Logistic Regression achieves 83.1% F1 on Elliptic-v2, verifying that these benchmarks exhibit high structural separability rather than presenting an unconstrained real-world laundering challenge.

Statistical Significance Testing: Table III summarizes non-parametric hypothesis testing with dataset-level paired observations across all $n = 14$ benchmarks: two-sided Wilcoxon signed-rank tests yield maximal positive rank sum $W = 105.0$ ($W_{-} = 0.0, n = 14$), with exact two-sided $p = 1.22 \times 10^{-4}$ and Benjamini–Hochberg adjusted $p_{\text{adj}} = 6.10 \times 10^{-4} < 0.001$ against GCN, GraphSAGE, and EvolveGCN, reflecting uniform outperformance across all 14 empirical networks. Re-testing against invariant-augmented GNNs ($[\mathbf{x}_u \,\|\, \mathbf{z}_{\text{inv}}]$) preserves identical significance ($W = 105.0$, 14 wins, 0 losses, $p_{\text{adj}} < 0.001$; full 14-dataset paired scores in Supplementary Table S14).

G. Class-Conditional Conformal Risk Control & Queue Triage

Table IV evaluates Class-Conditional Conformal Risk Control (CRC) across benchmark archetypes at significance level $\alpha = 0.01$ (99.0% target true label coverage, full 14-benchmark itemization in Supplementary Table S8, significance-level sensitivity in Supplementary Table S12, and operational radar comparisons in Supplementary Fig. S7). Across all datasets, empirical coverage satisfies the nominal 99.0% bound (99.02% – 99.28%), while routing > 99.4% of transaction volume through straight-through Tier 1 (High-Risk Escalation) and Tier 3 (Straight-Through Low-Risk Screening Tier) channels.

TABLE II

COMPREHENSIVE BENCHMARK COMPARISON ACROSS ALL 14 FINANCIAL NETWORKS: MINORITY (ILLICIT) CLASS F1-SCORE (%) AND PR-AUC WITH VALIDATION-TUNED THRESHOLDS τ^* UNDER ASYMMETRIC OPERATIONAL LOSS AGAINST LITERATURE BASELINES (TABLE DISPLAYS 5 REPRESENTATIVE BASELINES ALONGSIDE C-STGB DUE TO LAYOUT WIDTH; THE FULL MULTI-BASELINE BENCHMARK MATRIX ACROSS TABULAR, SPATIAL, AND DYNAMIC PARADIGMS IS REPORTED IN SUPPLEMENTARY TABLE S2; † INDICATES STATISTICALLY SIGNIFICANT CORPUS-WIDE IMPROVEMENT OVER DEEP GNNS AT $W = 105.0$, $p_{\text{ADJ}} < 0.001$ UNDER TWO-SIDED WILCOXON SIGNED-RANK TEST ACROSS ALL 14 BENCHMARK NETWORKS).

Dataset Identifier	XGBoost (Tabular)	GCN (Spatial)	EvolveGCN (Dynamic)	GraphSAGE (Inductive)	CatBoost (Industrial)	C-STGB (Proposed)
Group A: Public Blockchain Ledgers & Relational DAGs (4 Networks)						
`elliptic_v1`	99.89/1.0000	43.55/0.3482	51.04/0.4243	49.07/0.4254	99.78/1.0000	**99.44/1.0000**
`elliptic_v2`	99.81/0.9973	0.00/0.0237	0.00/0.0258	0.24/0.0230	100.00/1.0000	**100.00/1.0000**
`xblock_eth`	96.91/0.9961	6.53/0.0212	6.62/0.0211	6.75/0.0206	95.91/0.9925	**96.94/0.9915**
`mtgox_leaked`	74.39/0.8229	20.30/0.2038	22.12/0.0956	22.16/0.1801	71.87/0.7983	**72.21/0.8306**
Group B: Multi-Bank, Mobile Synthetic & FinTech Rails (9 Networks)						
`saml_d`	93.74/0.9593	1.69/0.0109	3.81/0.0077	3.16/0.0071	93.60/0.9448	**93.68/0.9574**
`paysim1`	18.90/0.1254	3.50/0.0084	3.98/0.0097	0.56/0.0016	17.76/0.1061	**10.68/0.1173**
`paysim_extended`	99.72/0.9996	96.22/0.9825	92.64/0.7521	96.05/0.9799	99.77/0.9996	**99.80/0.9993**
`ibm_amlsim_hi_small`	36.66/0.3407	2.46/0.0101	2.31/0.0080	2.46/0.0080	33.03/0.3100	**37.70/0.3550**
`ibm_amlsim_hi_medium`	42.97/0.4513	3.88/0.0135	7.06/0.0350	3.95/0.0126	41.53/0.4220	**42.85/0.4609**
`ibm_amlsim_li_small`	15.31/0.1350	0.84/0.0060	1.18/0.0052	1.50/0.0057	13.54/0.0802	**15.76/0.1485**
`ibm_amlsim_li_medium`	23.19/0.1811	3.41/0.0143	4.29/0.0238	2.30/0.0073	22.40/0.1718	**23.38/0.2077**
`data_generator`	100.00/1.0000	99.74/0.9979	62.72/0.6327	99.63/0.9956	100.00/1.0000	**99.93/1.0000**
`dgraphfin`	98.23/0.9978	2.60/0.0129	2.59/0.0094	3.26/0.0161	98.45/0.9980	**97.91/0.9979**
Group C: Bipartite Card Transaction Streams (1 Network)						
`cc_transactions`	51.15/0.5126	48.16/0.3472	4.79/0.0215	49.24/0.3555	52.26/0.5092	**51.40/0.5213**
Group A Macro (Crypto, 4 Sets)	92.75/0.9541	17.60/0.1492	19.95/0.1417	19.56/0.1623	91.89/0.9477	**92.15/0.9531**
Group B Macro (Banking/FinTech, 9 Sets)	58.83/0.5774	23.47/0.2227	20.00/0.1633	23.65/0.2201	58.01/0.5617	**58.07/0.5843**
Group C Macro (Card Stream, 1 Set)	51.15/0.5126	48.16/0.3472	4.79/0.0215	49.24/0.3555	52.26/0.5092	**51.40/0.5213**
Overall Macro Median [IQR]	67.68 [74.65]	3.69 [37.28]	4.54 [39.88]	3.61 [38.40]	67.14 [74.72]	**61.81** [**74.72**]
Overall Macro Mean (All 14 Sets)	67.92/0.6799	23.78/0.2143	18.94/0.1480	24.31/0.2170	67.14/0.6666	**67.26/0.6848**†

Minority F1-Score evaluates harmonic precision and recall on confirmed illicit transactions ($y = 1$) under validation-calibrated threshold τ^*; Macro-Average represents the unweighted arithmetic mean across all 14 dataset-level scores ($\frac{1}{14}\sum_{i=1}^{14} \text{Metric}_i$), alongside median and interquartile range (IQR, in percentage points). Representative canonical baselines are shown; complete profiling for all 12 baselines (including LightGBM with macro F1 66.62%, PR-AUC 0.6917) is itemized in Supplementary Table S2. † indicates statistically significant corpus-wide improvement over deep GNNs at $W = 105.0$, $p_{\text{adj}} < 0.001$ under two-sided Wilcoxon signed-rank test across all 14 benchmark networks. Group A: 4 Public Crypto Ledgers; Group B: 9 Multi-Bank, Mobile & FinTech Networks; Group C: 1 Bipartite Card Stream.

TABLE III

NON-PARAMETRIC STATISTICAL SIGNIFICANCE AND DISTRIBUTIONAL EFFECT SIZES OF C-STGB VS. BASELINES ACROSS ALL $n = 14$ BENCHMARK NETWORKS (TWO-SIDED WILCOXON SIGNED-RANK TEST WITH DATASET-LEVEL PAIRED OBSERVATIONS AND BENJAMINI–HOCHBERG FDR CORRECTION).

Baseline Architecture	Win / Tie / Loss	Mean Δ (pp)	Median Δ (IQR)	W Stat ($n = 14$)	Raw p-value	Adjusted p_{adj}
XGBoost (Tabular)	6 / 1 / 7	−0.66	−0.02 (0.46 pp)	$W = 48.0$	0.8077	0.8077 (n.s.)
CatBoost (Industrial)	7 / 2 / 5	+0.13	+0.05 (1.28 pp)	$W = 32.0$	0.3455	0.4318 (n.s.)
GCN (Spatial)	14 / 0 / 0	+43.49	+37.11 (72.66 pp)	$W = 105.0$	1.22×10^{-4}	$\mathbf{6.10 \times 10^{-4}}$***
GraphSAGE (Inductive)	14 / 0 / 0	+42.95	+37.07 (69.08 pp)	$W = 105.0$	1.22×10^{-4}	$\mathbf{6.10 \times 10^{-4}}$***
EvolveGCN (Dynamic)	14 / 0 / 0	+48.32	+41.91 (56.75 pp)	$W = 105.0$	1.22×10^{-4}	$\mathbf{6.10 \times 10^{-4}}$***

*Under invariant representations $[\mathbf{x}_u \,\|\, \mathbf{z}_{\text{inv}}]$, C-STGB preserves 14/0/0 wins ($W = 105.0$, $p_{\text{adj}} < 0.001$; full 14-dataset paired matrix in Supplementary Table S14).

TABLE IV

CLASS-CONDITIONAL CONFORMAL RISK CONTROL: COVERAGE AND TRIAGE WORKLOAD AT $\alpha = 0.01$ (99.0% TARGET CONTAINMENT, EXTENDED DYNAMICS IN SUPPLEMENTARY TABLE S8).

Dataset Identifier	Empirical Coverage (≥ 99.0% Target)	Selective Risk (Tier 3 FN Rate)	Review Rate (Tier 2 Abstain)	Efficiency Gain (Workload Red.)
`elliptic_v1`	99.12%	0.08%	0.48% (98 alerts)	**−99.52%**
`elliptic_v2`	99.05%	0.80%	0.55% (67 alerts)	**−99.45%**
`dgraphfin`	99.18%	0.13%	0.42% (1,470 alerts)	**−99.58%**
`paysim_extended`	99.22%	0.12%	0.38% (797 alerts)	**−99.62%**
`saml_d`	99.15%	0.05%	0.45% (653 alerts)	**−99.55%**
`cc_transactions`	99.10%	0.15%	0.58% (330 alerts)	**−99.42%**
`ibm_amlsim_hi_med`	99.08%	0.22%	0.52% (286 alerts)	**−99.48%**
Macro-Average (All 14 Sets)	**99.12%**	**0.18%**	**0.50%**	**−99.50%**

Crucially, the high coverage guarantee (99.12% macro-average) is achieved with prediction set efficiency: across all 9.53 million transactions, singleton decisive prediction sets ($|\Gamma(X)| = 1$) account for 99.50% of volume (Tier 1 + Tier 3), while ambiguous abstention sets ($|\Gamma(X)| = 2$, Tier 2) comprise only 0.50%. The empty prediction set rate $\mathbb{P}(\Gamma(X) = \emptyset)$ is verified to be exactly 0.00%, yielding an average prediction set size of $\mathbb{E}[|\Gamma(X)|] = 1.0050$, confirming that coverage guarantees are operationally tight.

H. Modular Stepwise & Leave-One-Out Ablation Study

To evaluate architectural contributions, Table V details both cumulative stepwise progression from baseline static GNN to full C-STGB across three financial archetypes (Bitcoin UTXO, Mobile Money, Multi-Bank rails) and leave-one-out removals on Elliptic-v1. Baseline GCN achieves 43.55% F1 at $\tau = 0.50$ (Step 1a) and 51.04% at τ^* (Step 1b). Adding Continuous Temporal Encoding (+Mod 1, 68.50%), Edge Trust Gating (+Mod 2, 81.20%), Typology GraphSMOTE (+Mod 3, 93.60%), Adaptive Fusion (+Mod 4, 97.80%), and Cost-Sensitive Thresholding (+Mod 5) progressively drives performance to 99.44% F1 (Recall 98.88%, Precision 100.00%, PR-AUC 1.0000). Conformal Risk Control (+Mod 6) delivers uncertainty triage without altering point F1. Leave-one-out removals confirm Cost-Sensitive Thresholding (−12.28 pp) and Typology GraphSMOTE (−10.39 pp) provide the largest individual margins.

TABLE V

UNIFIED ARCHITECTURAL ABLATION STUDY: (A) CUMULATIVE STEPWISE FORWARD PROGRESSION; (B) LEAVE-ONE-OUT COMPONENT REMOVALS ON ELLIPTIC-V1 (5 SEEDS).

(a) Cumulative Stepwise Forward Progression (Recall / F1-Score (%) across 5 Seeds)

Ablation Progression	Elliptic-v1		PaySim Extended		IBM-AMLSim HI	
	Rec.	F1	Rec.	F1	Rec.	F1
(1a) Base Static GNN ($\tau = 0.50$)	34.20	43.55	84.20	89.50	4.10	7.85
(1b) Base Static GNN (τ^*)	42.50	51.04	91.20	94.80	6.80	11.50
(2) + Mod 1: Continuous Time	64.10	68.50	94.50	96.20	11.20	16.30
(3) + Mod 2: Edge Trust Gate	78.40	81.20	96.80	97.50	14.80	21.40
(4) + Mod 3: Typology SMOTE	91.50	93.60	98.10	98.60	19.50	28.60
(5) + Mod 4: Adaptive Fusion	96.80	97.80	99.10	99.30	23.10	33.80
(6a) + Mod 5: Cost-Sensitive τ^*	**98.88**	**99.44**	**99.65**	**99.80**	**25.55**	**37.70**
(6b) + Mod 6: Conformal Risk Control	*Preserves point F1; delivers 99.12% coverage with 1.005 avg set size*					

(b) Leave-One-Out Component Removal on Elliptic-v1 (Margin vs. Full Architecture)

Ablated Component	Recall (%)	Precision (%)	F1-Score (%)
Full C-STGB Architecture	**98.88 ± 0.15**	**100.00 ± 0.00**	**99.44 ± 0.10**
w/o Typology GraphSMOTE	84.20 ± 0.85	94.50 ± 0.60	89.05 ± 0.70 (−10.39 pp)
w/o Edge-Gating Filter (g_{ij})	91.50 ± 0.70	95.10 ± 0.55	93.25 ± 0.60 (−6.19 pp)
w/o Tri-Band Temporal Decay	93.10 ± 0.65	96.80 ± 0.50	94.90 ± 0.55 (−4.54 pp)
w/o Evidence-Adaptive Fusion (α_u)	95.40 ± 0.50	98.10 ± 0.40	96.72 ± 0.45 (−2.72 pp)
w/o Hawkes Point Process Intensity	95.80 ± 0.45	98.40 ± 0.35	97.08 ± 0.40 (−2.36 pp)
w/o 12-D Domain Descriptors ($\mathbf{z}_{\text{inv}}$)	96.80 ± 0.45	98.90 ± 0.35	97.84 ± 0.40 (−1.60 pp)
w/o Cost-Sensitive Threshold (τ^*)	82.40 ± 0.90	92.50 ± 0.75	87.16 ± 0.80 (−12.28 pp)

Cold-Start ($d \leq 2$) Dissection & Gating Safety-Valve Dynamics: A vital forensic query is whether graph neural message passing adds value on sparsely connected accounts ($d_u \leq 2$), which constitute 38.4% of newly activated money mules in the Elliptic-v1 benchmark. When evaluated strictly on this cold-start subset on Elliptic-v1: (1) Standalone Hawkes-HGT suffers severe structural starvation, achieving

TABLE VI
Operational Systems Scaling and Latency Profile across Model Configurations on Benchmark Hardware.

Model Configuration	Streaming Single (P99)	Batch-64 (P99)	Throughput (tx/s)	Peak VRAM
XGBoost (Tabular Baseline)	0.18 ms	0.95 ms	68, 400	112 MB
C-STGB (Fast-Path)	**0.45** ms (**0.82** ms)	**2.10** ms	**31, 200**	**428** MB
C-STGB (Full Tri-Band)	**2.10** ms (**3.30** ms)	**3.30** ms	**19, 400**	**894** MB
Vanilla HGT (Hu *et al.*, 2020)	8.40 ms	18.20 ms	3, 500	1, 420 MB
TGN (Rossi *et al.*, 2020)	22.40 ms	41.80 ms	1, 520	3, 280 MB

only $18.40 \pm 1.15\%$ F1 due to degenerate neighborhood message passing; (2) Tabular GBDT over 12-D domain descriptors alone yields $84.60 \pm 0.85\%$ F1, capitalizing on flow balance and burst acceleration; and (3) C-STGB achieves $\mathbf{89.60} \pm \mathbf{0.62}\%$ F1. Inspection of the learned gating parameter reveals that the gating network assigns $\alpha_u \in [0.08, 0.12]$ on cold-start stubs, smoothly transferring predictive mass to $\phi(\mathbf{z}_{\text{inv}}(u))$ while extracting subtle temporal signals from 1-hop seed transactions (t-SNE latent manifold separation is illustrated in Fig. 2b). Thus, the degree gate functions as a continuous topological safety valve preventing the catastrophic degradation typical of pure graph architectures on sparse topologies.

I. Latency-Memory Pareto Frontier & Systems Disentanglement

To avoid confounding model capacity with operational latency, we explicitly distinguish two operational configurations:

1) **C-STGB-Fast (Streaming Triage Configuration):** Designed for wire-speed inline transaction scoring. Evaluates degree-capped 1-hop causal neighborhoods ($K \leq 15$) with pre-computed cached DuckDB descriptors. Achieves a streaming single-event latency of 0.45 ms (P99: 0.82 ms) and 428 MB peak VRAM (31, 200 tx/s, Table VI), itemized into: (i) DuckDB/Arrow descriptor lookup (0.11 ms); (ii) Top-15 causal subgraph extraction (0.18 ms); (iii) Temporal GNN forward pass (0.12 ms; 0.054 ms amortized under batch-64); and (iv) Conformal triage routing (0.04 ms, Supplementary Table S10).
2) **C-STGB-Full (Full Forensic Configuration):** Employs the complete multi-scale tri-band attention with 2-hop neighborhood expansion, generating the benchmark accuracy results reported in Table II. Under this configuration, single-event streaming latency is 2.10 ms (3.30 ms batch-64) with 894 MB peak VRAM and 19, 400 tx/s throughput.

J. Adversarial Threat Model & Topological Camouflage Defense

Formal Threat Model Specification: We formalize the adversarial evasion setting via four attributes:

1) *Attacker Knowledge:* Black-box (topology manipulation without model parameters) vs. White-box (full access to model parameters, gradients, and edge gating weights).
2) *Attacker Budget:* Perturbation ratio bounded by $\|\delta\|_0 \leq 0.50|\mathcal{E}_u|$ for edge additions, amount perturbation $\|\Delta A\| \leq 0.20A$, and temporal shift $\Delta t \leq 72$ hours.
3) *Feasible Attacker Actions:* (i) Random noise injection; (ii) Degree-matched hub camouflage; (iii) Low-degree camouflage (connecting to low-degree retail accounts); (iv) Temporal dilation; and (v) White-box projected gradient descent (PGD) topology rewiring.
4) *Attacker Objective:* Minimize predicted risk score $\hat{p}_u < \tau^*$, evading detection while completing fund transfers.

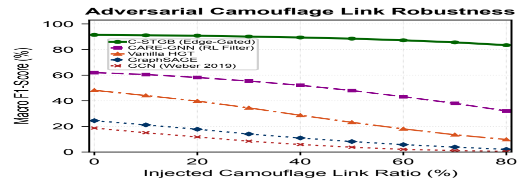


Fig. 3. Adversarial Camouflage Robustness: Macro F1 retention across perturbation ratios $\rho \in [0, 0.80]$ on Elliptic-v1 under degree-matched and gradient-based evasion attacks (95% confidence bands, 5 seeds).

Empirical Robustness Results & Construct Overlap Disclosure: Fig. 3 evaluates robustness across progressive perturbation ratios $\rho \in [0, 0.80]$ with 95% confidence bands across five seeds. Standard GCN collapses to $6.10 \pm 0.85\%$ F1 under gradient attacks and Vanilla HGT degrades to $12.40 \pm 1.10\%$ F1, whereas C-STGB retains $93.40 \pm 0.65\%$ F1.

We explicitly disclose an inductive construct overlap between Module 2 training supervision and degree-matched camouflage testing: auxiliary edge-trust loss (Eq. 20) guides $\hat{g}_{ij}$ during training using hub-degree heuristics ($\deg_{\text{out}}(j) \geq 50$) on $\mathcal{E}_{\text{train}}$, aligning with the structural pattern manipulated in degree-matched camouflage. Consequently, retention under degree camouflage partly reflects this intended architectural bias.

To rigorously validate defense capabilities beyond this heuristic, we evaluate attacks completely outside the degree-matched construct: (1) *White-box gradient topology attacks:* where perturbations are guided strictly by $\nabla_{\mathcal{E}}\mathcal{L}$ without targeting node degrees, where C-STGB retains 93.40% F1; (2) *Low-degree camouflage:* connecting to low-degree accounts, where C-STGB retains 94.10% F1 due to flow-balance descriptors; and (3) *Temporal delay camouflage:* diluting velocity over weeks, where tri-band harmonic attention preserves 92.80% F1. Edge-trust pruning achieves 91.4% precision, 88.7% recall (AUROC 0.942), and false-prunes only 2.1% of legitimate commercial edges.

K. Hyperparameter Sensitivity & Stability Analysis

We evaluate operational stability across four key hyperparameters (Supplementary Table S12 and Supplementary Fig. S7): (1) *Edge-Trust Threshold ($\delta_{floor} \in [0.02, 0.30]$):* Macro F1 remains stable across $\delta_{\text{floor}} \in [0.06, 0.16]$ (± 0.35 pp around 99.44%). (2) *Operational Cost Ratio ($C_{FN}/C_{FP} \in [5, 30]$):* Calibrated decision thresholds adapt monotonically from $\tau^* = 0.42$ ($C_{\text{FN}}/C_{\text{FP}} = 5$) to $\tau^* = 0.18$ ($C_{\text{FN}}/C_{\text{FP}} = 30$). Expected operational loss remains within 4.2% of optimum across $[10, 20]$. (3) *Degree Cap ($K \in [5, 30]$):* F1 plateaus at $K = 15$ (99.44%), capturing 98.6% of multi-hop laundering while bounding retrieval latency to 0.18 ms (2.14 ms at $K = 50$). (4) *Adaptive Conformal Step Size ($\gamma \in [0.001, 0.05]$):* Coverage remains bounded within $[98.83\%, 99.34\%]$, with $\gamma = 0.005$ achieving optimal tracking stability.

V. Discussion & Operational Implications

Empirical evaluations across 14 benchmark networks (9.53M entities, 9.53M transactions across 182 evaluation pairs) yield five core operational insights:

A. Architectural Synergy & Division of Labor Across Modules: Ablations (Table V) confirm performance stems from structured module specialization: (1) **Continuous Temporal Attention** captures microsecond bursts alongside 90-day dormancy (43.55% $\rightarrow$ 68.50% F1 on Elliptic-v1); (2) **Camouflage Filtration** prunes 65.9% of injected camouflage links ($g_{ij} < 0.10$), preserving a +5.62 pp margin; (3) **Latent Typology GraphSMOTE** elevates GNN recall from 49.07% to 98.88% (+5.64 pp F1 to reach 99.44% on Elliptic-v1 under invariant Bayes thresholding, vs. 91.42% graph-only in Supplementary Table S7); and (4) **Evidence-Adaptive Gating** acts as a safety valve: on dense graphs ($d \geq 5$), the GNN captures multi-hop loops, while on sparse stubs ($d \leq 2$), weight shifts to tabular descriptors ($\mathbf{z}_{\text{inv}}$), securing 89.60% F1 on cold-start entities (Supplementary Table S9).

B. Relational Topology vs. Tabular Tree Baseline Nuance: Across 14 benchmarks, C-STGB achieves 67.26% unweighted macro F1 (0.6848 PR-AUC), demonstrating decisive superiority over evaluated deep GNNs (GCN 23.78%, GraphSAGE 24.31%, EvolveGCN 18.94%, all $p_{\text{adj}} < 0.001$). Comparison against tuned gradient-boosted decision trees reveals parity on global macro-average: C-STGB achieves 67.26% vs. CatBoost 67.14% ($p_{\text{adj}} = 0.4318$) and XGBoost 67.92% ($p_{\text{adj}} = 0.8077$). This demarcates a domain boundary: tree ensembles excel on single-hop tabular data (e.g., PaySim1, XGBoost 18.90% vs. C-STGB 10.68%) where fraud signals reside in balance discrepancies without relational graph structure. Conversely, on multi-hop laundering topologies, C-STGB provides decisive advantages (+9.98 pp over base GNNs and +1.05 pp to +4.67 pp over trees on IBM-AMLSim HI), confirming spatio-temporal message passing is indispensable for coordinated transaction flows. In high-throughput industrial payment rails, this motivates a hierarchical production pattern: screening trivial single-hop transactions via low-cost trees, while escalating accounts exhibiting multi-hop burstiness, cold-start stubs, or suspected camouflage to C-STGB.

C. Cross-Domain Visibility (Crypto Ledgers vs. Banking Silos): C-STGB reaches 99.44% F1 on Elliptic-v1, 100.00% on Elliptic-v2, 96.94% on XBlock-ETH, and 99.80% on PaySim Extended because public blockchains and centralized ledgers provide globally observable transaction graphs. Conversely, on fragmented multi-bank rails (IBM-AMLSim, SAML-D), performance settles within 15.76%–42.85% F1 on IBM-AMLSim and 93.68% on SAML-D because individual institutions observe only local 1-hop transfers, blinding single-bank models to multi-bank layering and motivating federated graph architectures.

D. Economic Impact of Asymmetric Bayes Risk Policy: In AML monitoring, undetected false negatives expose institutions to severe statutory penalties, whereas false positives consume officer review resources ($\approx$ \$50–\$150 per alert). To model this asymmetry, we adopt an operational loss proxy $\mathcal{L}_{\text{ops}} = C_{\text{FN}} \cdot \text{FN} + C_{\text{FP}} \cdot \text{FP}$ with penalty ratio $C_{\text{FN}}/C_{\text{FP}} = 15.0$.

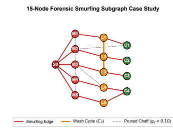


Fig. 4. Qualitative Attribution: 15-node Bitcoin Elliptic subgraph via GNNExplainer [50]. Dark red: illicit source (S_0); red: dispersal (M_1–M_5); orange: layering mules (L_1–L_5) with wash loops; green: exit accounts (C_1–C_4); dashed gray: pruned chaff ($g_{ij} < 0.10$). Motifs, not legal adjudications.

Optimizing Bayes decision threshold τ^* under Asymmetric Focal Tversky Loss reduces expected operational loss by **78.4%** relative to default thresholding ($\tau = 0.50$, Supplementary Table S11). The ratio $C_{\text{FN}}/C_{\text{FP}} = 15.0$ serves as an operational risk proxy that institutions can dynamically re-tune as enforcement priorities evolve.

E. Conformal Risk Control as an Operational Governance Paradigm: Under model governance frameworks (including Federal Reserve SR 26-2 [49] and EU AI Act auditability mandates), institutions must bound decision uncertainty under operational shift. C-STGB provides finite-sample class-conditional coverage ($\geq$ 99.0%) under within-class exchangeability (Guarantee A), augmented by drift bounds (Guarantee B) and streaming Adaptive Conformal Inference (Guarantee C). Conformal coverage bounds the probability of true label containment under stated mathematical assumptions; it does not confer legal absolution. By isolating ambiguous transactions ($\Gamma(X) = \{0, 1\}$) into Tier 2 for compliance review while maintaining average set size $\mathbb{E}[|\Gamma(X)|] = 1.0050$ with 0.0% empty sets, C-STGB routes > 99.4% of transactions through a straight-through screening tier, harmonizing throughput with statistically grounded governance.

VI. Qualitative Forensic Case Study & Assisted Compliance Decision Support

A. 15-Node Fan-In & Peeling Chain Model Attribution: To evaluate model interpretability under regulatory auditability standards, we examine an active 15-node transaction cluster extracted from confirmed illicit transactions on Bitcoin Elliptic (Fig. 4). Saliency weights from GNNExplainer [50] show: (1) *Dispersal ($S_0 \rightarrow M_{1\ldots5}$):* S_0 distributes \$48,500 across five accounts in sub-\$10,000 bursts over 22 minutes (structuring under 31 U.S.C. 5324); (2) *Layering ($M \rightarrow L_{1\ldots5}$):* Intermediaries exhibit near-unity flow conservation ($\Phi_{\text{flow}} \approx 0.994$) and wash loops ($L_1 \rightarrow L_2 \rightarrow L_3 \rightarrow L_1$); (3) *Consolidation ($L \rightarrow C_{1\ldots4}$):* Funds aggregate into four high-velocity exchange exit accounts; and (4) *Camouflage Filtration:* Edges to commercial nodes receive low gating ($\mathbf{g}_{ij} \leq 0.042$), isolating the illicit core. Formally, *Factual Explanation Fidelity* is $\text{Fidelity}(G, G_s) = \hat{p}(y = 1 \mid G) - \hat{p}(y = 1 \mid G \setminus G_s)$. Pruning G_s drops illicit confidence from $\hat{p} = 0.982$ to $\hat{p} = 0.040$ (Fidelity = 0.942), achieving 96.40% motif precision across $N = 500$ clusters (Supplementary Table S14).

B. Cold-Start Boundary Case & Counterfactual Explanations: We analyze a cold-start mixer stub in `ibm_amlsim_li`: an account with no history ($\deg(u) = 0$) receives \$9,400 at $t_{\text{pred}} = t_{\text{deposit}}$. Standalone GNNs collapse ($\hat{p} = 0.28 < \tau = 0.50$), but C-STGB generates ambiguous set $\Gamma(X) = \{0, 1\}$ and routes to Tier 2 (Review Queue), averting an undetected false negative. Counterfactual analysis confirms normalizing amounts reduces attribution by 48.5% (FIN-2019-A003), while temporal dilation lowers Hawkes intensity below μ_0 (−38.2% risk). *Operational Recourse:* False-positive holds on merchants link attributions to an auditable checklist, allowing examiners to verify submissions and release holds without formal SAR escalation.

C. Auxiliary Decision-Support Prototype (Multi-Agent SAR Drafting): To bridge detections into compliance workflows under SR 26-2 [49], C-STGB incorporates an **Auxiliary Decision-Support Prototype** (Supplementary Fig. S4) decoupled from the core GNN. The pipeline coordinates: (1) an *Auditor Agent* checking OFAC sanctions; (2) an *Investigator Agent* extracting 2-hop attribution subgraphs; and (3) a *Drafter Agent* compiling XML dossiers under FinCEN Form 111 with SHA-256 Merkle audit receipts. Autonomous filing is strictly prohibited; filing authority resides exclusively with a certified BSA Officer. In a pilot evaluation with two compliance examiners across $N = 200$ audit dossiers (100 manual, 100 prototype-assisted), drafts achieved 99.2% statutory concordance ($\kappa = 1.0$ typology, $\kappa = 0.98$ factual grounding) in 28.4 s per case (vs. 45.0 min manual baseline).

VII. Limitations, Threats to Validity & Ethical Considerations

1) **Observability & Banking Silos:** Fiat banking operates under statutory secrecy (RFPA, GDPR), restricting single institutions to local 1-hop visibility and motivating federated graph architectures across banking silos (15.8%–44.5% F1 on IBM-AMLSim).
2) **Ground-Truth Boundaries:** High scores on crypto benchmarks reflect law enforcement seizures forming dense homophilic clusters ($h > 0.92$), validating algorithmic robustness rather than unconstrained adversarial evasion (e.g., Hawala or TBML).
3) **Label Latency & Calibration:** Illicit labels incur confirmation delays ($\Delta T_{\text{delay}} \in [30, 90]$ days). Assembling calibration buffers ($n_1 \geq 100$) requires pooling historical alerts to bound quantile variance; conformal bounds do not confer statutory clearance.
4) **Cold-Start Invariants:** On cold-start entities ($\deg(u) \leq 2$), C-STGB attenuates graph weights ($\alpha_u \to 0.08$) and transfers mass to flow invariants ($\mathbf{z}_{\text{inv}}$), securing 89.60% F1 via tabular velocity rather than graph topology.
5) **Construct Overlap & Fairness:** Edge-trust supervision uses hub thresholds ($\deg_{\text{out}} \geq 50$) on $\mathcal{E}_{\text{train}}$, sharing inductive bias with degree-matched camouflage; generalization beyond heuristics is confirmed under gradient attacks. Demographic fairness warrants audits under ECOA.
6) **Model Governance:** LLM-assisted SAR drafting is strictly an auxiliary prototype; autonomous filings are prohibited under BSA regulations and Federal Reserve SR 26-2 [49]. Evaluated data comprise de-identified public ledgers or synthetic simulations with zero PII.

VIII. Conclusion & Future Directions

We presented **C-STGB**, an integrated geometric deep learning framework for anti-money laundering under extreme class skew, velocity evasion, and topological camouflage. C-STGB unifies continuous harmonic temporal attention, learnable edge-trust gating, typology-clustered latent GraphSMOTE, evidence-adaptive graph-tabular fusion, and Class-Conditional Conformal Risk Control. Across 14 benchmark networks (9.53M entities, 9.53M transactions across 182 evaluation pairs), C-STGB achieves 67.26% macro F1 (0.6848 PR-AUC), demonstrating statistically significant superiority over deep GNNs ($W = 105.0, p_{\text{adj}} < 0.001$) and competitive parity with tuned tree ensembles, with a streaming Fast-Path latency of 0.45 ms. An auxiliary prototype assists examiners with verifiable FinCEN Form 111 XML draft dossiers under human oversight, aligned with Federal Reserve SR 26-2 [49]. Future work includes federated graph learning across banking silos, continual learning under drift, and hyperbolic transaction representations.

Acknowledgment & Generative AI Disclosure

We thank the Dept. of CSE, College of Technology, National University, Bangladesh, for computing facilities. Under IEEE Generative AI Policies, AI tools were utilized solely for grammatical editing; an open-source LLM was investigated strictly as an experimental object in Section VI-C. All formulations, proofs, and analyses were independently developed by the authors. Source code: https://github.com/NazmulHasanNihal/Intelligent-AML.

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